BJT Current Source Experiment 1

ID & Name:

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Pre-LAB Work

1. For the circuit given in Fig. 1, find and expression for the current flowing through the R7 resistor and for the output resistance Ro of the circuit. Ignore the Early Effect and assume that all the transistors are operating in active mode. (Hint: To calculate the current you do not need to know the value of the R7 resistor. To calculate Ro, you better review the common base configuration.)

There are 3 current mirror subcircuit

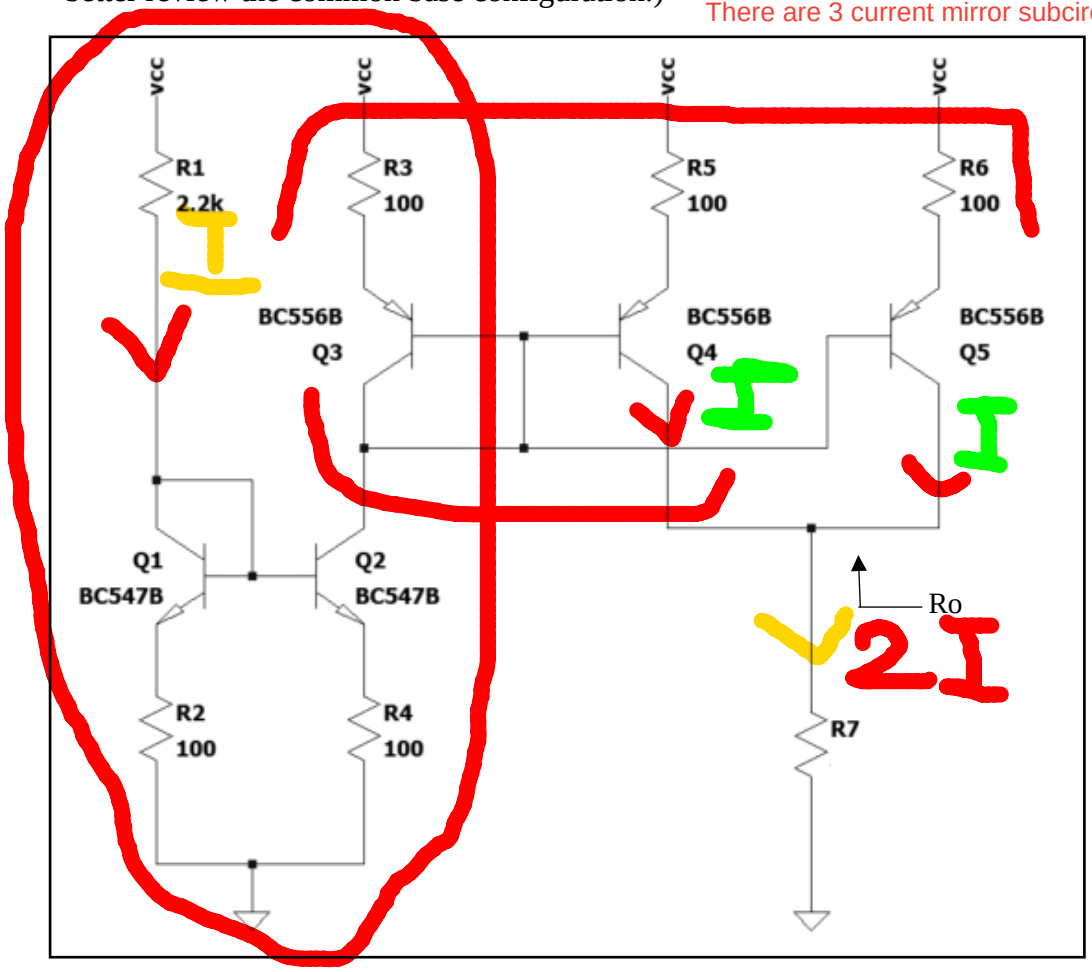


Figure 1

Answer:

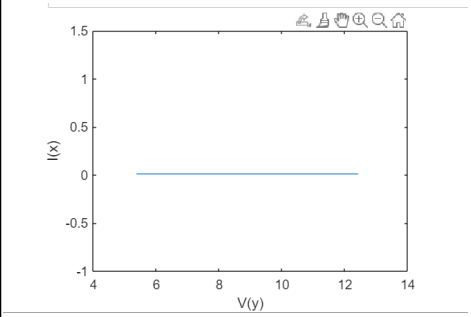
2. Let Vcc be 15V and simulate the circuit in Fig. 1 with SPICE for the R7 values listed in Table 1. Measure the voltage across the R7 resistor and the current through it and fill the table.

Table 1: Simulation Results

R7	**V (R7)**	**Ix = I (R7)**	**Vy = Vcc – |V (R7)|**
220Ω	2.583V	11.77mA	12.42V
330Ω	3.8744V	11.77mA	11.130V
750Ω	8.79V	11.77mA	6.21V
820Ω	9.627	11.77mA	5.37
2.2kΩ	14.534V	6.6mA	0.47V

3. Based on the simulation results in Table 1 (ignore the last row) plot Ix (x-axis) vs Vy (y-axis) and fit a line. Calculate the slope of the line which will be the output resistance Ro.

Plot:



As you can see it's sloop is zero as expected.

Equipment/Component List

Table 3: Equipment/Component List

Equipment/Component	Model/Value	Count
Breadboard	-	1
Resistor	2.2kΩ	**2**
Resistor	100Ω	**5**
Resistor	220Ω	1
Resistor	330Ω	1
Resistor	750Ω	1
Resistor	820Ω	1
BJT (npn)	BC547	**2**
BJT (pnp)	BC556	**3**
Multi-meter	-	1
Voltage Source	15V	1
Probes & Cables	-	-

Experiment Procedure

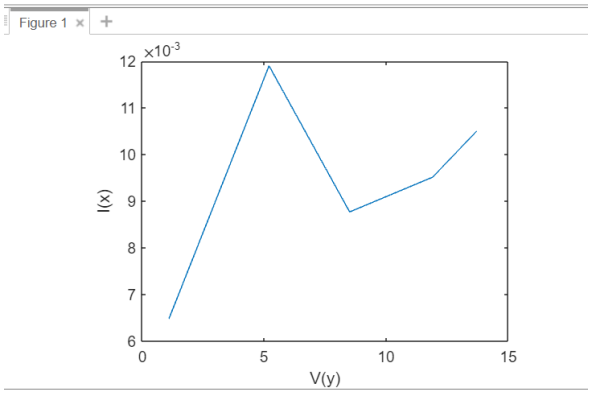
4. Set-up the circuit in Fig. 1 with the resistor values on the Fig. 1. Adjust Vcc to 15V. Values for the R7 resistor are given in Table 2. Measure the exact values of R7 resistors before connecting them to the circuit and fill the second column of the table. Connect the R7 resistor to the circuit, measure the voltage across it and fill the table.

Table 2

R7	Measured (R7)	V (R7)	Ix = |V (R7)| / Meas. (R7)	Vy = Vcc – |V (R7)|
220Ω	217.9	2.3	10.56mA	13.7V
330Ω	325.8	3.1	9.51mA	11.9
750Ω	741	6.5	8.77mA	8.5
820Ω	816.9	9.8	11.99mA	5.2
2.2kΩ	2.17k	14.1	6.49mA	1.1

5. Based on the data in Table 2 (ignore the last row), plot Ix (x-axis) vs Vy (y-axis) and fit a line. Calculate the slope of the line which will be the output resistance Ro.

Plot:



6. Compare the measured quantities in Table 2 with your simulation results in Table 1 and calculations in part 1. When R7 resistor is set to 2.2k the current significantly changes. Comment on the reason of this change.

Answer:

Because of the technical problems in the lab we could not get the result even close to supposed to be ,we have tried a few times but all the results were ridiculous.I m sure we are going to do much better in next lab.
When the R7 is 2.2k ohm output resistance changes significantly as a result of this gain is decreased so current is also decreased.

